



# Cortical EEG slowing in resting-state Parkinson's disease

Final presentation: model, code, and findings

PD EEG group · Rui, Jan, Melissa, Friedrich · 26.06.2026

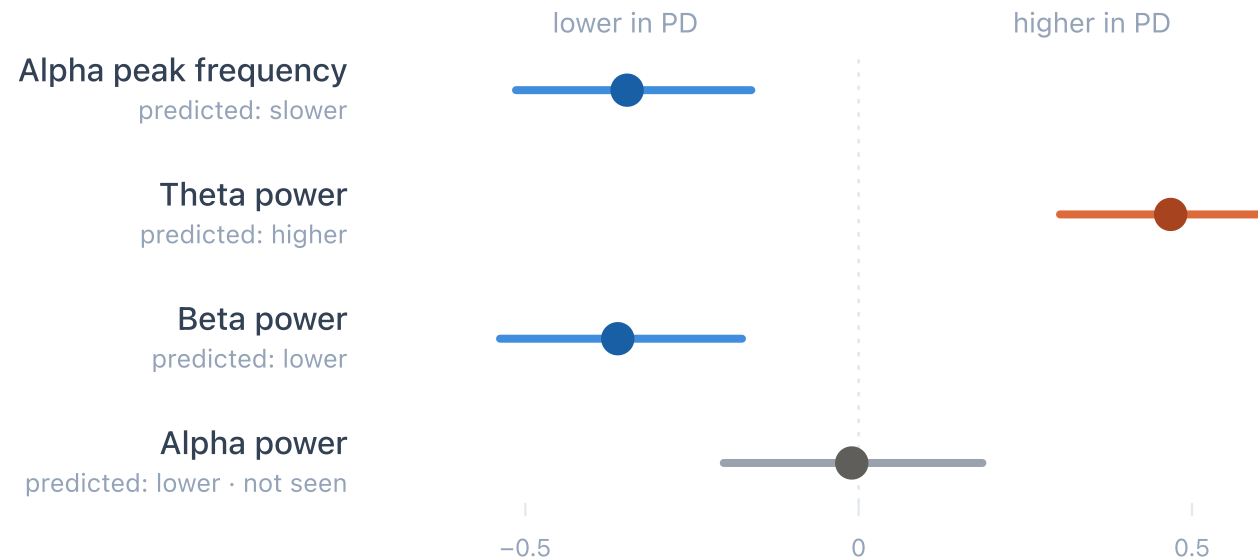
# Hypotheses

Compared to age-matched healthy controls, Parkinson's patients at rest show:

- **slowing of the alpha peak frequency** (8 to 13 Hz)
- **increased theta-band power** (4 to 8 Hz)
- **decreased cortical beta-band power** (13 to 30 Hz)

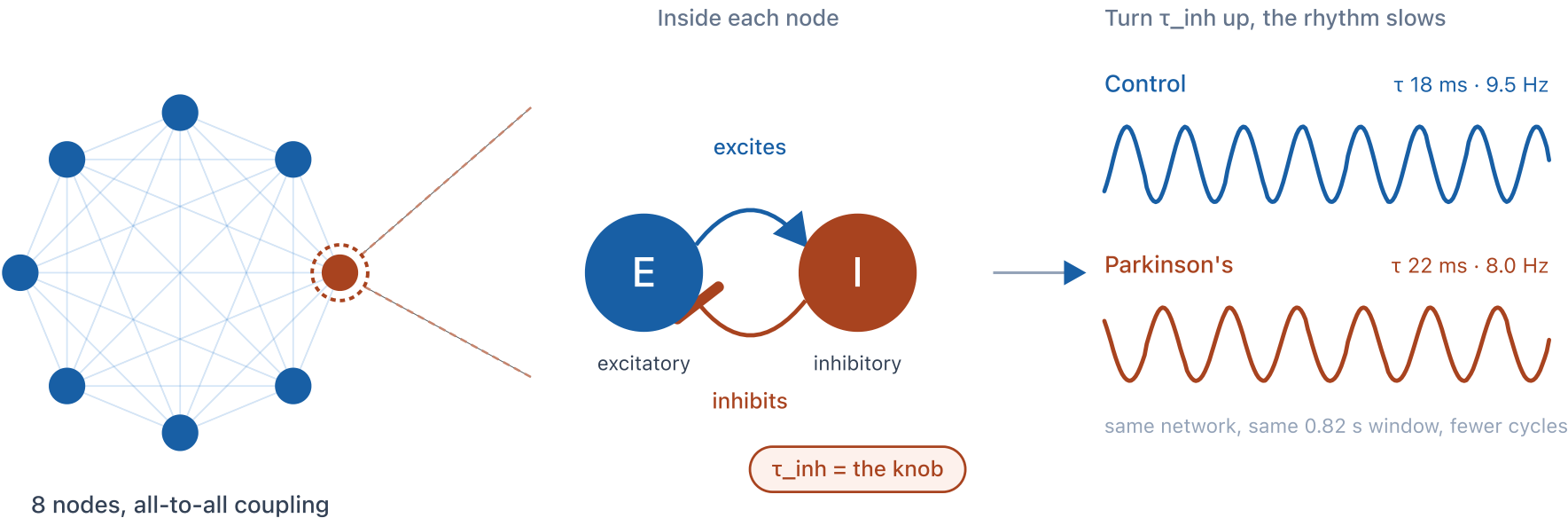
## How Parkinson's changes the resting EEG

effect size per feature, with 95% confidence interval



# The model: a Wilson-Cowan network

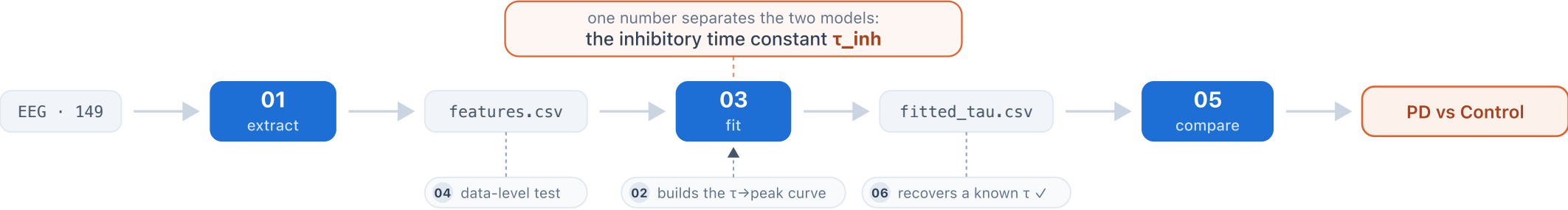
8 coupled cortical nodes, each an excitatory-inhibitory loop.  $\tau_{inh}$  is the only knob we turn.



The final adaptation from healthy to patient is a single parameter: a longer inhibitory time constant  $\tau_{inh}$ , about **+3.25 ms** in PD.

# Code & repository

THE PIPELINE EEG to result, through six numbered scripts



THE REPOSITORY

pd-eeg

main

01 ... 06, numbered

scripts/

- 01\_extract\_features.py
- 02\_simulate\_network.py
- 03\_fit\_per\_subject.py
- 04\_compare\_features.py
- 05\_compare\_tau.py
- 06\_recover\_tau.py

src/

results/

- fitted\_tau.csv
- tau\_recovery.csv

figures/

slides/

features · model · fitting · stats

every run writes here

compare\_tau.png · tau\_recovery.png

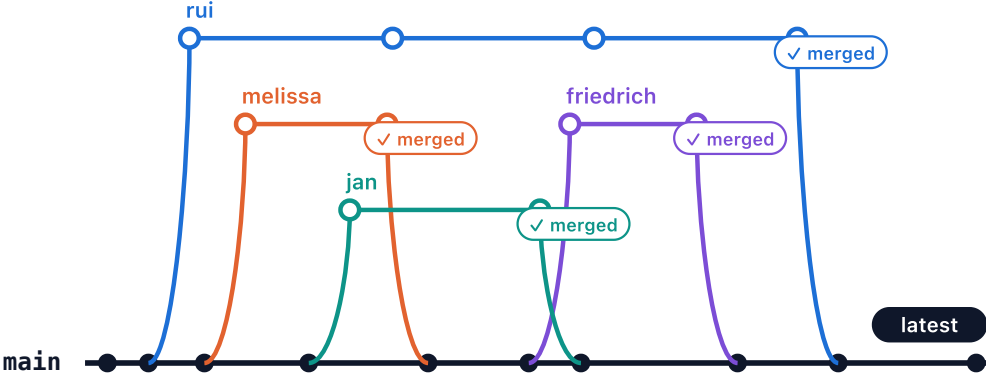
decks + notes

every result is one script run

```
bash
$ python scripts/05_compare_tau.py
Control 18.25 → PD 21.50 ms diff +3.25 p = 0.00025 ***
→ wrote results/figures/compare_tau.png
$ python scripts/06_recover_tau.py
22 known tau recovered bias -0.10 max err 1.75 ms
→ wrote results/tau_recovery.csv
```

VERSION CONTROL

feature branches, reviewed, merged into main



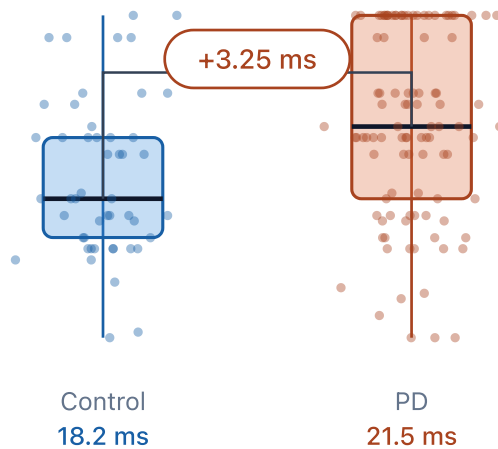
46 commits · feature branches → reviewed → merged into main · fully reproducible

# Does the model separate patients from controls?

## Fitted inhibitory time constant

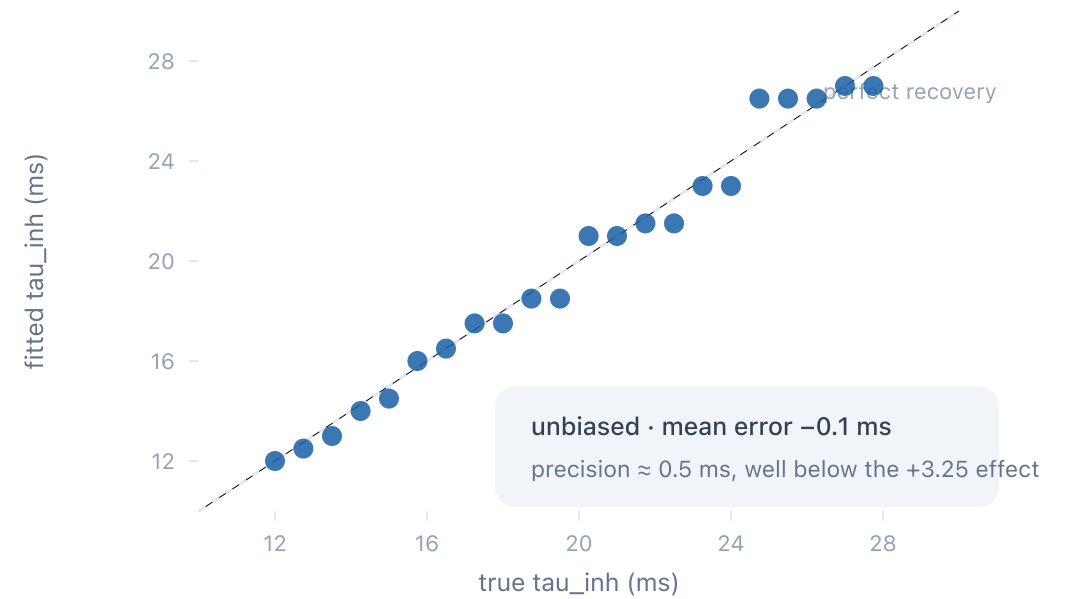
the model parameter we change for a patient

fitting recovers tau to  $\pm 0.5$  ms, far below this group gap



## Validation: recovering a known parameter

simulate at a known tau, fit it back



Fitted  $\tau_{inh}$  is **+3.25 ms** higher in PD (one-sided Mann-Whitney,  $p \approx 0.0002$ ). The recovery test shows this is a real separation, not an artefact of the fitting.

# Outlook and open challenges

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## CURRENT CHALLENGES

- One parameter ( $\tau_{inh}$ ) reproduces the alpha slowing, but not the theta and beta changes
- The fit re-expresses the data-level effect, rather than adding independent evidence

## FUTURE DIRECTIONS

- Fit more than the peak (full spectrum or functional connectivity), on a real connectome
- Relate fitted  $\tau$  to clinical scores and to known inhibitory (GABAergic) changes in PD

# What we took away

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**Rui** [your biggest learning]

**Jan** [your biggest learning]

**Melissa** [your biggest learning]

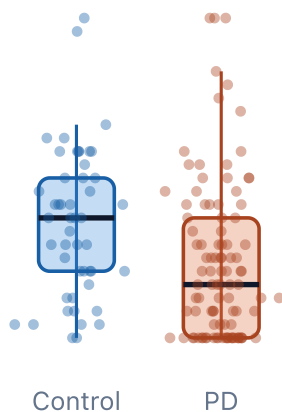
**Friedrich** [your biggest learning]

# Appendix: per-subject distributions

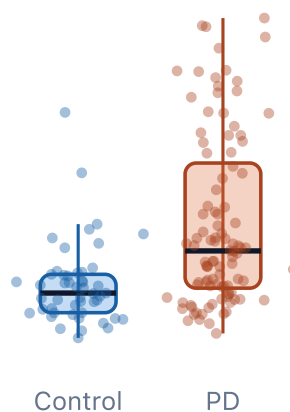
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## PD vs Control: per-subject EEG features

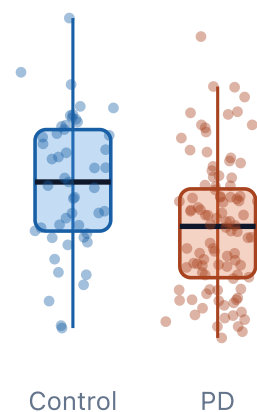
Alpha peak frequency  
slower in PD



Theta power  
higher in PD



Beta power  
lower in PD



Alpha power  
unchanged

